MARITIME COMPLIANCE TOOLS

USER GUIDE

NavStab — Ship Stability & Resistance Calculator Suite

Holtrop–Mennen Resistance · Stability · Trim & Drafts · Knowledge Hub

Prepared for: Vessel Compliance & Reporting Teams

Document version: 1.0

Table of Contents

1 Introduction

1.1 Purpose of this guide

This guide explains how to use NavStab, a browser-based Naval Architecture and Ship Stability platform. It walks through the dashboard, the full Calculators Suite (11 tools covering hull-form coefficients, stability, trim, hydrostatics and resistance), the dedicated Holtrop–Mennen Resistance calculator, and the Knowledge Hub theory reference — using screenshots taken directly from the tool.

The guide is written for naval architects, ship stability officers, marine engineering students and technical superintendents who need to enter hull-form and loading data, run calculations, and interpret the results.

1.2 What the tool does

- Calculates hull-form design coefficients (Cb, Cw, Cp, Cm), displacement and waterplane area from principal dimensions.
- Computes transverse stability parameters — KB, BM, KM, GM — for box, triangular and ship-shaped hulls, plus moments of weights, angle of loll, inclining-experiment analysis and heel while turning.
- Calculates trim and new drafts from MCTC, LCF and weight-shift data, plus dock-water density and Fresh Water Allowance (FWA).
- Provides TPC & Displacement, Bilging, Free Surface Effect, Drydocking, Rolling Period and Ship Squat calculators for day-to-day loading and seakeeping checks.
- Runs the Holtrop–Mennen (1982/1984) statistical regression method to predict calm-water resistance (RF, RW, RT), the form factor (1+k1) and effective power (PE) across a speed range.
- Includes a Knowledge Hub with ten short theory modules covering the fundamentals behind every calculator.

Before you start

- The tool runs entirely in your browser — no data is uploaded or sent anywhere.
- Have the vessel's principal particulars, hydrostatic data and loading condition figures ready before opening a calculator.
- Use a current version of Chrome, Edge, or Firefox for the best experience.
- Results are not saved between sessions — record or screenshot important figures before closing or reloading the page.

2 Interface Overview

NavStab opens on a single-page dashboard. A fixed header bar sits above the content at all times, and the dashboard body holds the hero panel, the Calculators Suite grid, and a shortcut into the Knowledge Hub.

Figure 1 — Header toolbar: Back, Home and User Guide controls.

Control	Function
← Back	Returns to the previous browser page. If no previous page exists, it returns to the dashboard.
⌂ Home	Returns directly to the main NavStab dashboard from anywhere in the platform.
User Guide (?)	Opens the tool's built-in on-screen guide, listing every dashboard option, calculator and Knowledge Hub topic.
Back to Dashboard	Shown inside every calculator page; returns from that calculator to the main dashboard.

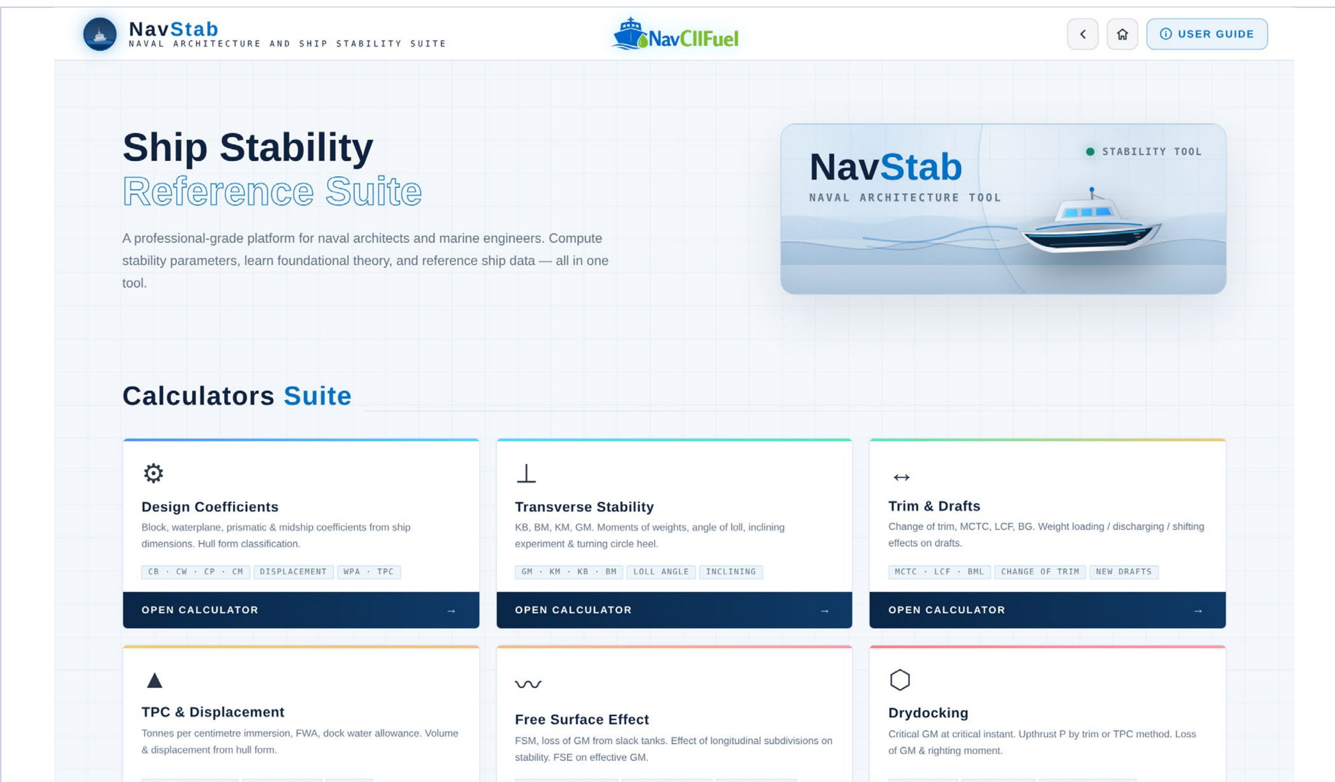


Figure 2 — Dashboard hero panel with the NavStab logo mark.

Scrolling down the dashboard reveals the Calculators Suite grid — a set of clickable cards, one per calculator — followed by a shortcut banner into the Knowledge Hub Stability Theory module.

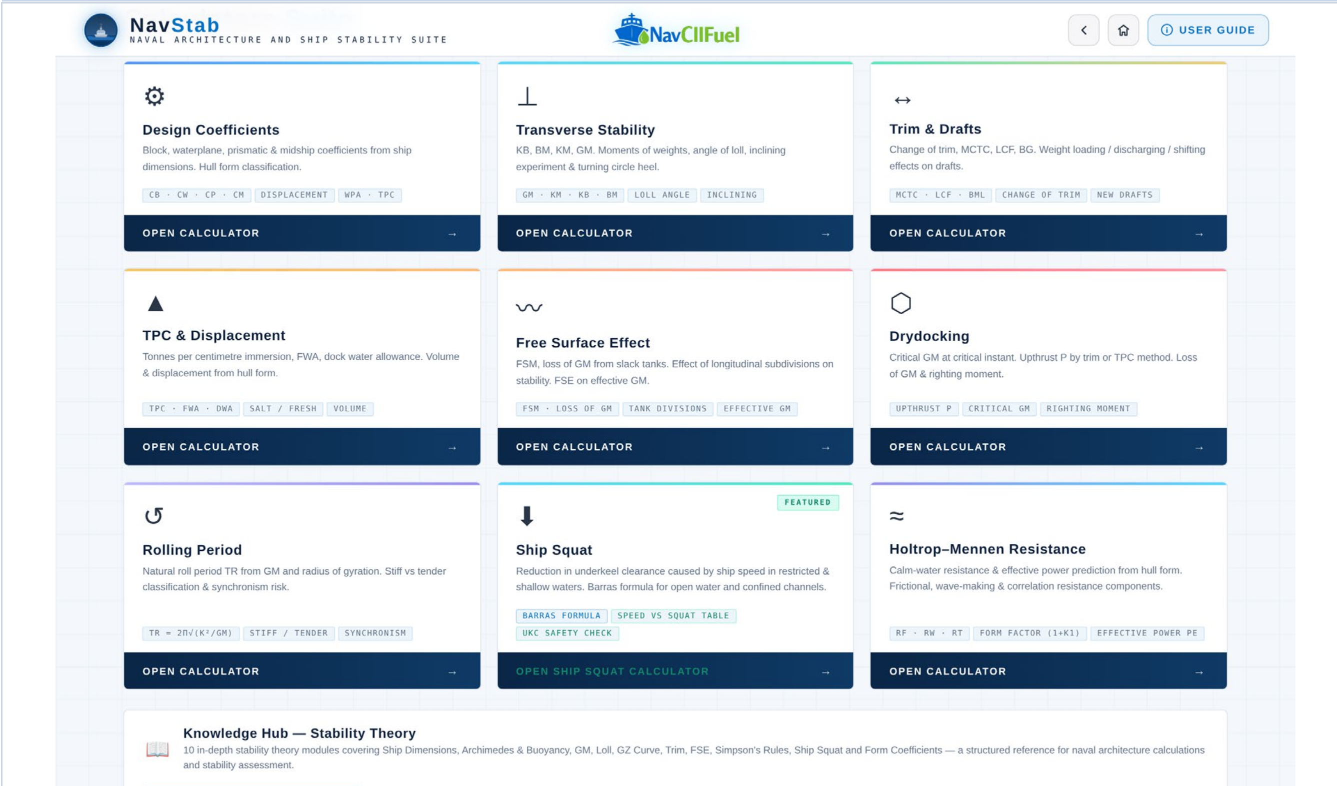


Figure 3 — Calculators Suite grid on the dashboard.

3 Quick Start Workflow

Follow these six steps for a typical single-calculation session:

Step	What to do
1	From the dashboard, scroll to the Calculators Suite and click the card for the calculation you need (e.g. Design Coefficients, Transverse Stability, Holtrop–Mennen Resistance).
2	Enter the vessel's principal dimensions and hull-form data into the input fields shown on the left-hand card. Units are shown beside each field label.
3	Where a calculator has multiple related tools (e.g. Transverse Stability, Trim & Drafts), use the sub-tabs near the top of the page to switch between them.
4	Click the primary action button (e.g. Calculate, Calculate Resistance, Calculate New KG) to run the calculation.
5	Review the results card, chart or speed/resistance table that appears alongside or below the input card.
6	Use Clear to reset a calculator's fields, or Back to Dashboard to return and open a different tool.

4 Calculators Suite — Full Reference

The table below summarises every calculator available from the dashboard grid. Each opens as its own page with a Back to Dashboard button.

Calculator	What it calculates
Design Coefficients	Block, waterplane, prismatic and midship coefficients (Cb, Cw, Cp, Cm), displacement and waterplane area, from LBP, breadth, draft, displacement and midship area.
Transverse Stability	KB, BM and KM/GM for box, triangular or ship-shaped hulls; plus four sub-tools — Moment of Weights, Angle of Loll, Inclining Experiment and Angle While Turning.
Trim & Drafts	Change of trim, MCTC, LCF and BG effects on drafts from loading, discharging or shifting weight; plus Density and Fresh Water Allowance (FWA) sub-tools.
TPC & Displacement	Tonnes per centimetre immersion from waterplane

Calculator	What it calculates
	area or from length/breadth/Cw, and displacement/volume from draft and Cb.
Free Surface Effect	Free surface moment (FSM) and loss of effective GM from slack tanks, including the effect of longitudinal subdivisions.
Drydocking	Upthrust P by the trim or TPC method, critical GM at the critical instant, and loss of righting moment during docking.
Rolling Period	Natural roll period TR from GM and radius of gyration, with stiff/tender classification and synchronous-rolling risk indication.
Ship Squat	Sinkage and reduction in under-keel clearance caused by vessel speed in open or restricted/shallow water, using the Barras formula.
Holtrop–Mennen Resistance	Calm-water resistance and effective-power prediction (RF, RW, RT, form factor, PE) from hull-form particulars, across a speed range.

Note

- Bilging (compartment flooding / lost buoyancy) is available as a supporting calculation alongside TPC & Displacement.
- Ship Squat is marked FEATURED on the dashboard and includes a speed-vs-squat reference table.

5 Using a Calculator — Design Coefficients Example

Section A of the Design Coefficients calculator captures the vessel's principal particulars. This walkthrough shows a complete single-calculator workflow that applies, with minor variation, to every tool in the suite.

Field	Description
Length Between Perpendiculars — LBP (m)	The vessel's length between perpendiculars, used in the block and prismatic coefficient formulae.
Breadth Moulded — B (m)	Moulded breadth, used in the block, waterplane and midship coefficient formulae.
Draft Moulded — d (m)	Moulded draft at the condition being assessed.
Displacement — W (tonnes)	Displacement at the given draft, used to derive the block coefficient.
Midship Area (m²)	Immersed midship section area, used to derive the midship coefficient Cm.
Water Density ρ (t/m³)	Defaults to 1.025 t/m³ (salt water); change for fresh or brackish water calculations.

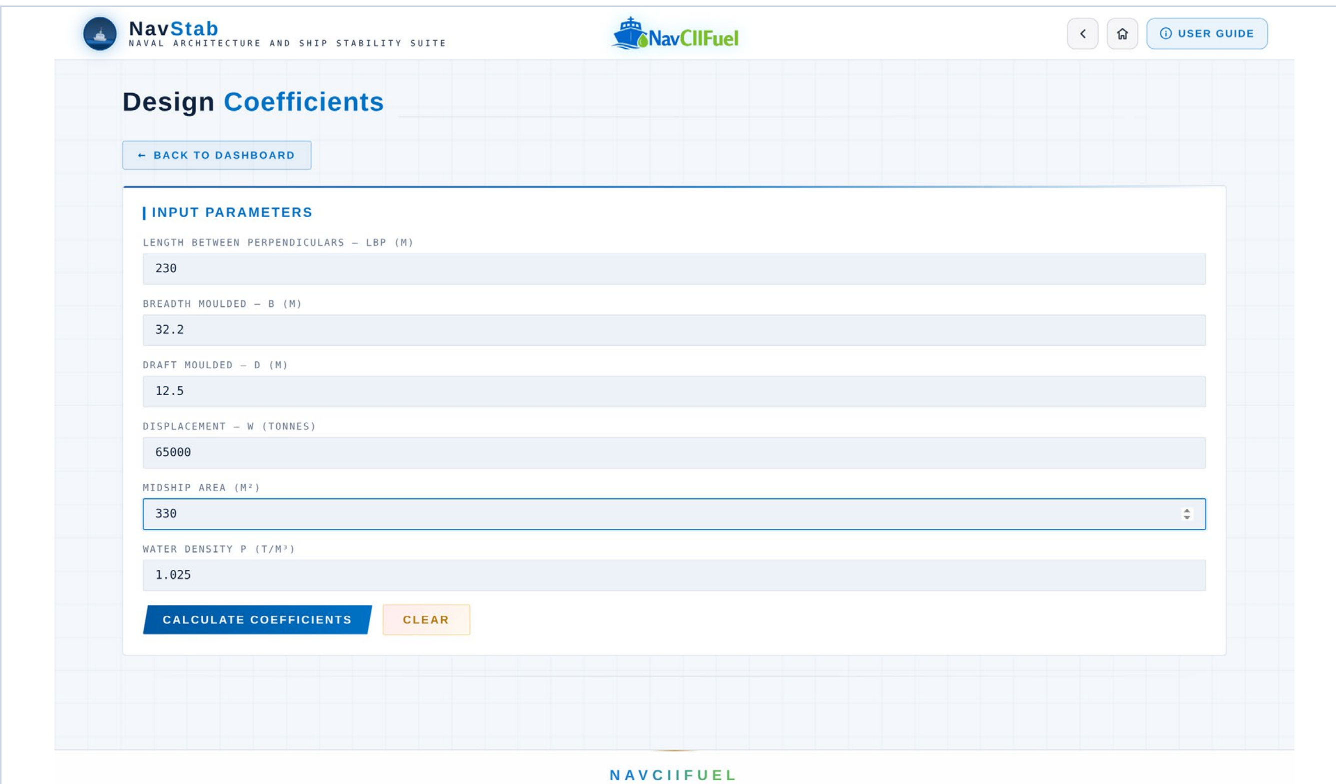


Figure 4 — Design Coefficients calculator with sample vessel data entered.

Click Calculate Coefficients to compute Cb, Cw, Cp and Cm from the entered particulars. The page also reveals a Typical Cb Values reference table, letting you compare the calculated block coefficient against typical values for ULCCs, tankers, bulk carriers, general cargo ships, passenger liners, container ships and tugs. Use Clear to reset all fields and start again.

6 Transverse Stability Module

Transverse Stability opens with five sub-tabs, each addressing a different stability calculation. Use the sub-tabs near the top of the page to switch tools without leaving the module.

Sub-tab	Purpose
GM Calculator	KB, BM and KM/GM for box-shaped, triangular or ship-shaped hulls, from draft, breadth, length, KG and (for ship-shaped hulls) Cb and Cw.
Moment of Weights	New KG and GM after loading, discharging or shifting weights, using a running moments table.
Angle of Loll	Loll angle from a negative GM and BM — applies only when GM is negative.
Inclining Experiment	GM and KG derived from an inclining test: displacement, KM, shifted weight, distance shifted, plumb-line length and mean deflection.
Angle While Turning	Heel angle while turning from speed, turning-circle radius, BG and GMT.

Figure 5 — Sub-tabs at the top of the Transverse Stability page.

Example — GM Calculator: select the vessel shape (Box-shaped, Triangular or Ship-shaped), enter draft, breadth, length and KG, and — for a ship-shaped hull — the block and waterplane coefficients. Click Calculate to display KB, BM, KM and GM in the results card.

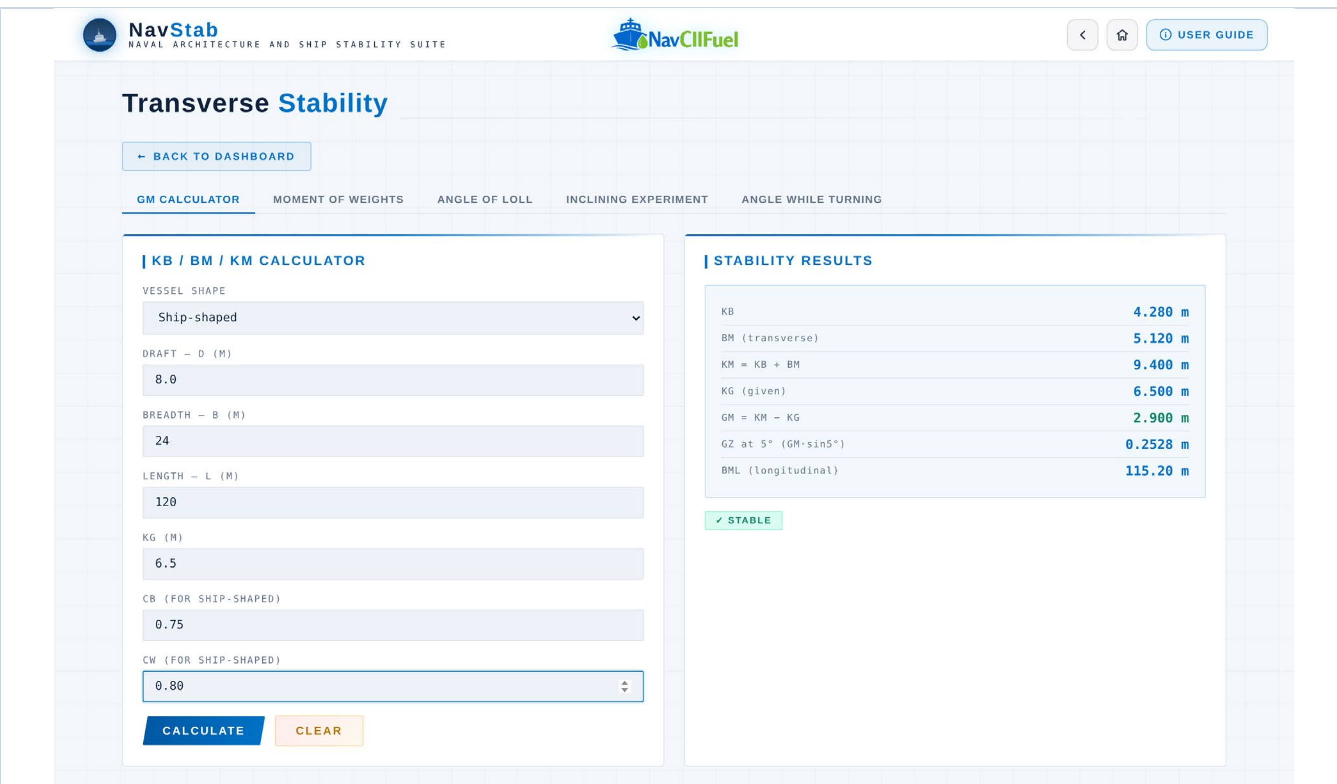


Figure 6 — GM Calculator with a ship-shaped hull calculated.

Tip

- Cb and Cw are only used for the Ship-shaped option — they are ignored for Box-shaped and Triangular hulls.
- A negative GM result signals a lolled or unstable condition; follow up with the Angle of Loll sub-tab.

7 Trim & Drafts Module

Trim & Drafts calculates the change of trim and new drafts fore and aft after loading, discharging or shifting a weight, using LBP, LCF, MCT 1cm and TPC. The module also includes Density and Fresh Water Allowance (FWA) sub-tools for moving between salt, dock and fresh water.

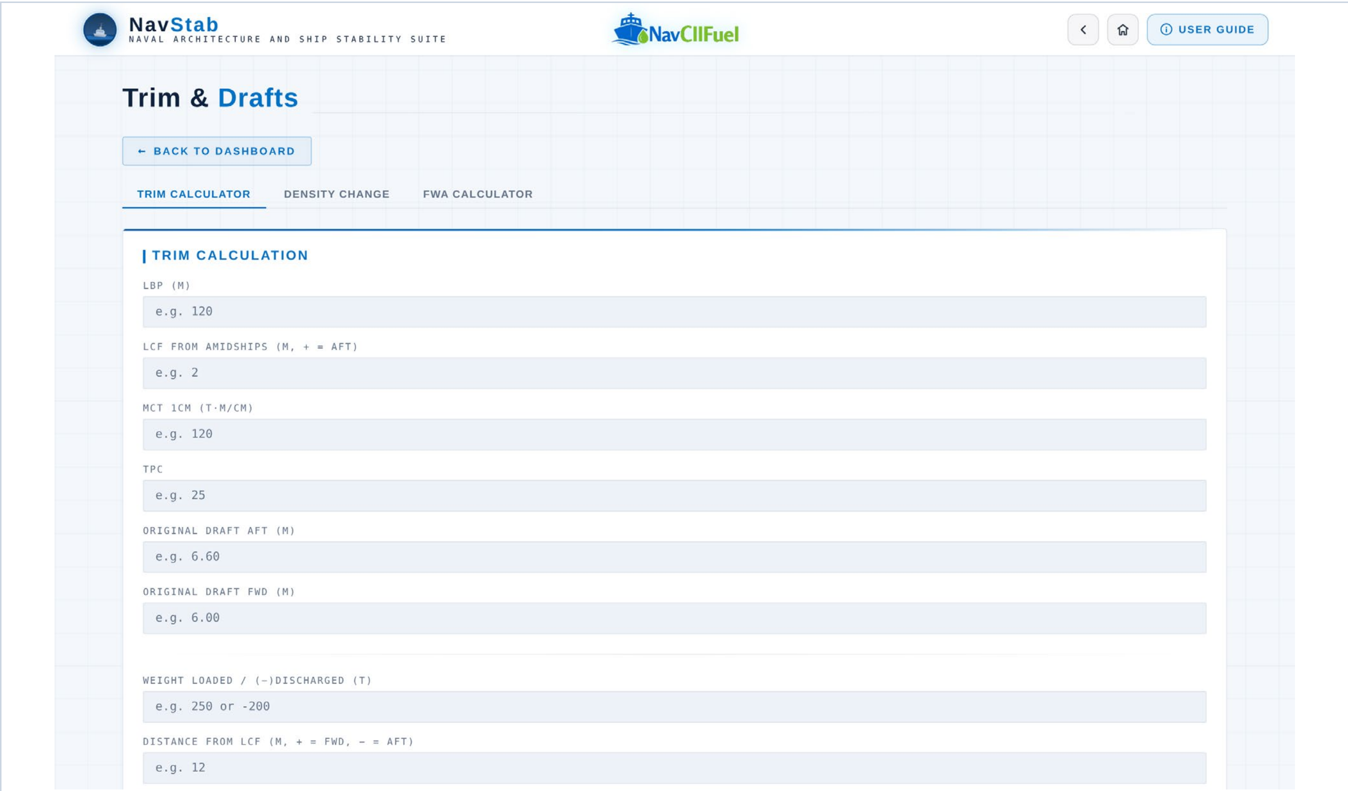


Figure 7 — Trim & Drafts calculator layout.

Field	Description
LBP (m)	Length between perpendiculars, used to apportion the change of trim to the fore and aft drafts.
LCF from amidships (m)	Longitudinal position of the centre of flotation; positive values are aft of amidships.
MCT 1cm (t·m/cm)	Moment to change trim one centimetre.
TPC	Tonnes per centimetre immersion, used to find the parallel rise/sinkage from the weight change.
Original Draft Aft / Fwd (m)	The vessel's drafts before the weight change.
Weight Loaded / (−)Discharged (t)	Enter a positive value for loading, negative for discharging.
Distance from LCF (m)	Longitudinal distance of the weight from LCF; positive is forward, negative is aft.

The Density sub-tool calculates new drafts when moving between water densities (e.g. salt to fresh water), and the FWA sub-tool derives the Fresh Water Allowance from summer displacement and salt-water TPC.

8 Holtrop–Mennen Resistance Calculator

The Holtrop–Mennen Resistance calculator applies the Holtrop–Mennen (1982/1984) statistical regression method to estimate calm-water resistance and effective power from principal hull-form parameters. It assumes no bulbous bow and no immersed transom. Open it from the Calculators Suite grid on the dashboard.

Field	Description
Length on Waterline Lwl (m)	Waterline length, the primary length parameter for the regression.
Beam B (m)	Moulded beam.
Draft T (m)	Moulded draft at the design condition.
Block Coefficient Cb	Block coefficient at the design draft.
Midship Coefficient Cm (optional)	Auto-estimated from Cb if left blank.
Waterplane Coefficient Cwp (optional)	Auto-estimated from Cb if left blank.
LCB from midship, %L (optional)	Longitudinal centre of buoyancy; positive is forward of midship.
Appendage Wetted Area Sapp (m², optional)	Wetted surface of appendages (rudder, bilge keels, etc.); leave at 0 if unknown.
Design Speed Vk (knots)	Speed at which the resistance and power results are reported.

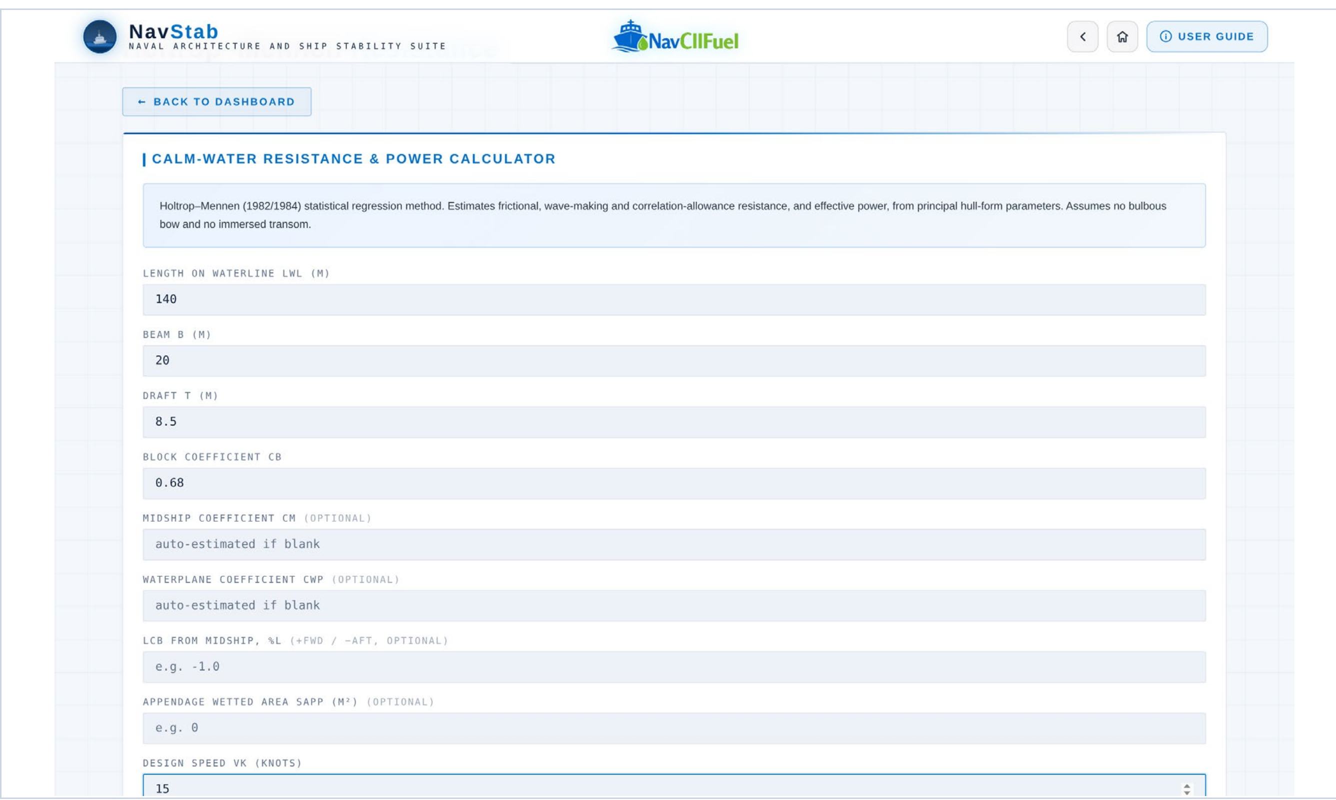


Figure 8 — Holtrop–Mennen calculator with sample hull-form data entered.

Click Calculate Resistance to generate the results. The results card reports the resistance components — frictional (RF), wave-making (RW) and total (RT) — together with the form factor (1+k1) and effective power (PE) at the design speed.

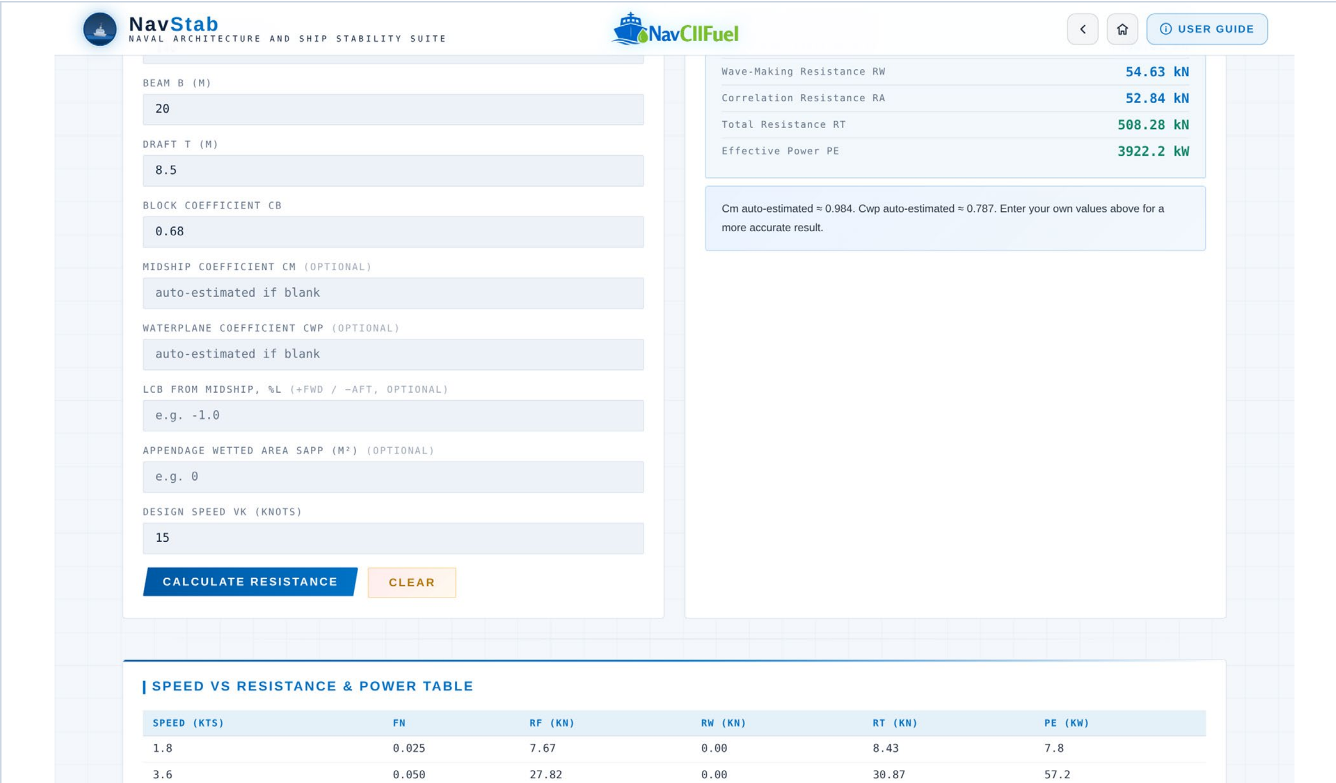
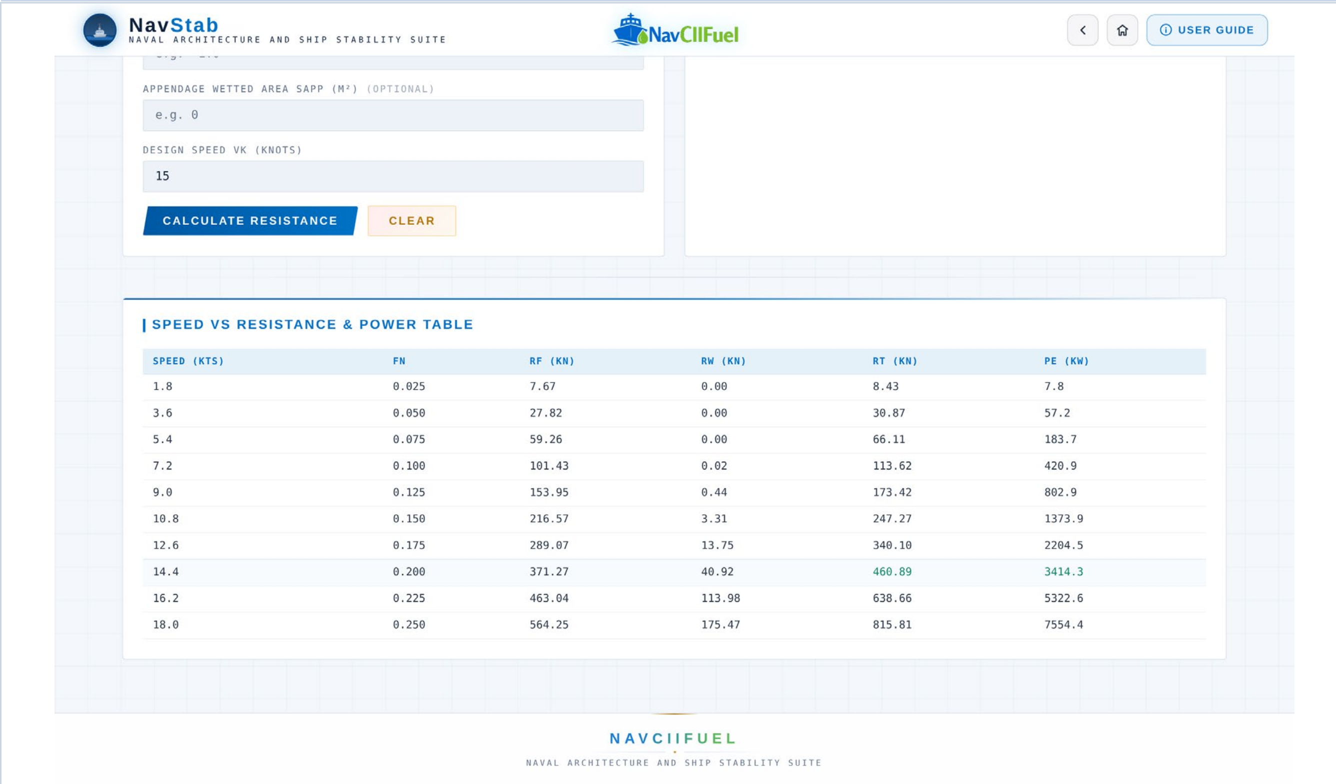


Figure 9 — Resistance and power results for the sample hull.

Below the results card, a Speed vs Resistance & Power table lists RT and PE across a range of speeds either side of the design speed, useful for early-stage speed-power trend review and comparison between design iterations.



SPEED (KTS)	FN	RF (KN)	RW (KN)	RT (KN)	PE (KW)
1.8	0.025	7.67	0.00	8.43	7.8
3.6	0.050	27.82	0.00	30.87	57.2
5.4	0.075	59.26	0.00	66.11	183.7
7.2	0.100	101.43	0.02	113.62	420.9
9.0	0.125	153.95	0.44	173.42	802.9
10.8	0.150	216.57	3.31	247.27	1373.9
12.6	0.175	289.07	13.75	340.10	2204.5
14.4	0.200	371.27	40.92	460.89	3414.3
16.2	0.225	463.04	113.98	638.66	5322.6
18.0	0.250	564.25	175.47	815.81	7554.4

Figure 10 — Speed vs Resistance & Power table.

Tip

- Leave Cm and Cwp blank to let the tool auto-estimate them from Cb — only override them if you have measured hull-form data.
- Because the method assumes no bulbous bow and no immersed transom, treat results for hulls with these features as indicative only.
- Use Back to Dashboard to return to the Calculators Suite grid.

9 Knowledge Hub — Stability Theory

The Knowledge Hub is reached from the dashboard shortcut banner below the Calculators Suite grid. It provides ten short theory modules that explain the concepts behind the calculators, useful as a quick reference or as training material.

Module	Covers
1. Ship Dimensions	Principal dimensions — length, breadth, depth, draft, freeboard — and their role in ship design and stability checks.
2. Archimedes & Buoyancy	Buoyant force, displacement, centre of buoyancy, and the principle that supports vessel flotation.
3. Metacentric Height (GM)	The relationship between KG, KM and GM, and why GM matters for initial transverse stability.
4. Angle of Loll vs List	The difference between an external heeling/list condition and an unstable loll caused by negative or very low GM.
5. GZ Curve	Righting lever, righting moment, range of stability, maximum GZ, and how the GZ curve is interpreted.
6. Trim	Longitudinal stability, change of trim, LCF, MCTC, and draft changes from shifting weights or loading changes.
7. Free Surface Effect	How liquid movement in slack tanks reduces effective GM, and why free-surface correction is required.
8. Simpson's Rules	Approximate area, volume and moment calculations used in hydrostatics and naval architecture.
9. Ship Squat	Sinkage and trim change due to speed in shallow/restricted water, and its importance for under-keel clearance.
10. Form Coefficients	Block, prismatic, waterplane and midship coefficients, and how they describe hull shape and fullness.

10 Best-Practice Checklist

- Double-check the unit shown beside each field (m, tonnes, t/m³, knots) before entering data — the tool does not convert units automatically.
- For the Design Coefficients and GM calculators, re-check inputs before reading results, since coefficients and stability values are highly sensitive to draft and breadth.
- Only override Cm and Cwp on the Holtrop–Mennen calculator when you have measured hull-form data; otherwise leave them blank for auto-estimation.
- Use consistent sign conventions: positive LCB/LCF is forward of midship, positive weight is loaded (negative is discharged), and GM is entered as negative for the Angle of Loll calculator.
- Record or screenshot results before navigating away or reloading — the tool does not save data between sessions.
- Treat all outputs as engineering support / training reference figures, not as a substitute for the vessel's approved stability booklet, loading computer or class-approved documentation.

11 Troubleshooting

Symptom	Likely cause / fix
Results panel stays empty	The primary action button (e.g. Calculate, Calculate Resistance) has not been clicked yet, or a required field is empty — fill all fields and click the button.
Ship-shaped GM result looks wrong	Cb and/or Cw were left blank — these two fields are required only for the Ship-shaped option in the GM Calculator.
Holtrop–Mennen power looks too low/high	Check that Lwl, B, T, Cb and Vk are all entered in the stated units (m, m, m, dimensionless, knots) — a mismatched unit is the most common cause.
Data disappeared after reloading the page	NavStab does not persist data between sessions — re-enter values, or keep a separate record of the vessel particulars you use most often.
Wrong sub-tool is shown	Use the sub-tabs at the top of Transverse Stability or Trim & Drafts to switch — each sub-tool has independent input fields.
Can't find a specific calculator	Return to the dashboard with Home, then scroll to the Calculators Suite grid, or use the calculator names in Section 4 of this guide.

This is a sensitive engineering context: always verify final vessel stability, loading, drydocking, squat and operational decisions against approved ship documents, class requirements and applicable regulations.